Managerial Economics (ARE) 106eh
University of California, Davis
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KEY—Practice Homework Problems for Chapter 2 and 3 (This is Practice HW 2)

Problem 1:
(a) The stochastic error term, e_i, and the residual error term, e_i-hat mean the same thing.

***False*. The residual e_i-hat is an approximation (i.e., an estimator) of the true error term, e_i.**

(b) The PRF gives the value of the dependent variable corresponding to each value of the dependent variable.

***False*. It gives the *mean* value of the dependent variable, given the values of the explanatory variables.**

In the PRF, or, the expected value of y given x:

$$E(y_i/x) = \beta_1 + \beta_2 x_i.$$

See graph later in this HW

(c) A linear regression model means a model linear in the variable.

False, see discussion later in this HW.

Problem 2:
What is the relationship between (a) β_1 and b_1, (b) β_2 and b_2, and (c) e_i and e_i-hat.

(*a*) b_1 is an estimator of B_1.
(*b*) b_2 is an estimator of B_2.
(*c*) e_i-hat is an estimator of e_i.

We never observe B_1, B_2, and *e*. Once we have a specific sample, we can obtain their estimates via b_1, b_2 and *e-hat*.

Problem 3:
Consider a simple regression in which the dependent variable MIM = mean income of males who are 18 years of age or older, in thousands of dollars. The explanatory variable PMHS = percent of males 18 or older who are high school graduates. The data consist of 51 observations on the 50 states plus the District of Columbia. Thus MIM and PMHS are "state averages." The estimated regression, along with the standard errors and t-statistics, is:

$$\hat{MIM} = (a) + 0.180PMHS$$

(se)	(2.174)	(b)
(t)	(1.257)	(5.574)

(a) What is the estimated equation intercept? Show your work.

$b_1 = t*se(b_1) = 1.257*2.174 = 2.733$

(b) What is the standard error of the estimated slope? Show your work.

$se(b_2) = b_2/t = 0.180/5.754 = 0.0313$

(c) State the economic interpretation of the estimated slope. Is this sign what you expected, *a priori*.

The estimated slope $b_2 = 0.18$ indicates that a 1% increase in males 18 and older, who are high school graduates, increases average income of those males by $180. The positive sign is expected; more education should lead to higher salaries.

(d) Construct a 99% confidence interval of the slope of this relationship.

t-critical with $\alpha/2 = 0.005$ (or, $1 - \alpha/2 = 0.995$) and df = 49; t-critical = 2.68.

CI: (0.096, 0.264)

(f) Test the hypothesis that the slope of the relationship is 0.2 against the alternative that it is not. What does this result mean?

$t = (0.180 - 0.2)/0.0313 = -0.639$

At either a 5% or 1% level we do not reject H_0. This result is compatible with the data. Thus, it is plausible that a 1% increase in males 18 and older who are high school graduates leads to an increase in average income for those males of $200. Non-rejection of H_0 means this claim is compatible with the sample data.

$$\widehat{MIM} = (a) + 0.180PMHS$$

(se)	(2.174)	(b)
(t)	(1.257)	(5.574)

(a) <u>what is (a)</u>

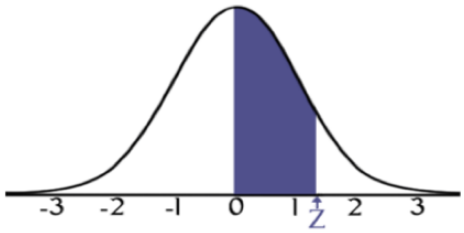
$$t = \frac{\text{estimate of } b_1 - \text{null value}}{\text{standard error of estimate of } b_1}$$

$$1.257 = \frac{(a) - 0}{2.174} \Rightarrow a = 1.257 * 2.174$$

<u>Remark</u> In general tables we are always reporting result of hypothesis testing of estimates with respect to 0.

	estimate	p-value	t-stat	se
b_1				
b_2				

$\Rightarrow$ referes $H_0: \beta_1 = 0$ vs $\beta_1 \neq 0$.

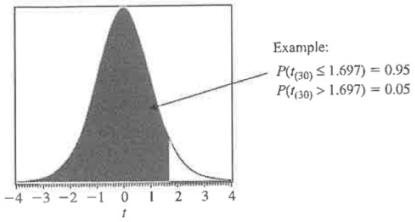


STANDARD NORMAL TABLE (Z)

Entries in the table give the area under the curve between the mean and *z* standard deviations above the mean. For example, for *z* = 1.25 the area under the curve between the mean (0) and *z* is 0.3944.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2969	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3513	0.3554	0.3577	0.3529	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998

HW2. problem 3.d.



Example:
$P(t_{(30)} \le 1.697) = 0.95$
$P(t_{(30)} > 1.697) = 0.05$

Table 2 **Percentiles of the *t*-distribution**

df	$t_{(0.90,df)}$	$t_{(0.95,df)}$	$t_{(0.975,df)}$	$t_{(0.99,df)}$	$t_{(0.995,df)}$
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
31	1.309	1.696	2.040	2.453	2.744
32	1.309	1.694	2.037	2.449	2.738
33	1.308	1.692	2.035	2.445	2.733
34	1.307	1.691	2.032	2.441	2.728
35	1.306	1.690	2.030	2.438	2.724
36	1.306	1.688	2.028	2.434	2.719
37	1.305	1.687	2.026	2.431	2.715
38	1.304	1.686	2.024	2.429	2.712
39	1.304	1.685	2.023	2.426	2.708
40	1.303	1.684	2.021	2.423	2.704
50	1.299	1.676	2.009	2.403	2.678
∞	1.282	1.645	1.960	2.326	2.576

Source: This table was generated using the SAS® function TINV.

df = 49. we are closer to 2.678 than 2.706. That's why in the problem we use 2.68 as the critical value.

Problem 4:

```
sample 1 63
read(are106/COAL.DAT) y x2 x3 x4 x5 x6 x7
stat / all
ols y x4
test x4 = 0.03
stop
```

Consider the SHAZAM command file (in the box) and the output file.

(a) You are to run the regression equation below on SHAZAM:

$$y_t = \beta_1 + \beta_2 x_{4t} + e_t.$$

State your values of b_1 and b_2, along with their respective t-statistics.

b_1 = 138.45; t_{b1} = 2.163
b_2 = 0.028967; t_{b2} = 9.217

(b) Construct a 95% confidence interval for β_2. Based on this confidence interval do you reject H_0: $\beta_2 = 0$?

The confidence interval has endpoints (0.022924; 0.0351). $\beta_2 = 0$ does not fall within this confidence interval and hence we reject H_0: $\beta_2 = 0$.

(c) Now use the t-statistic to conduct the same hypothesis text in part (h).

Assuming a two-tailed test ($\alpha = 0.05$ or $\alpha/2 = 0.025$) and 61 degrees of freedom, we find in the table that the critical t is 1.96. In addition, $t_{b2} = 9.217$. Thus, we reject H_0: $\beta_2 = 0$.

(d) Now redo parts (h) and (i) but now testing H_0: $\beta_2 = 0.03$.

In this case, $\beta_2 = 0.03$ falls within the confidence interval and we cannot reject H_0: $\beta_2 = 0.03$.

In addition, $t_{b2} = (0.028967 - 0.03)/0.003143 = -0.3287$. Again, we cannot reject H_0: $\beta_2 = 0.03$.

(e) What are the values for SST, SSR, and SSE?

SSR = $0.97670*10^7$; SSE = $0.70137*10^7$; SST = $0.16781*10^8$

(f) Verify that $R^2 = 0.5820$ by using the numbers of part (j).

R^2 = SSR/SST = $0.97670*10^7/0.16781*10^8 = 0.5820$

<u>Problem 4</u>—Shazam output.

SHAZAM OUTPUT

```
|_sample 1 63
|_read(are106/COAL.DAT) y x2 x3 x4 x5 x6 x7
UNIT 88 IS NOW ASSIGNED TO: are106/COAL.DAT
   7 VARIABLES AND       63 OBSERVATIONS STARTING AT OBS       1

|_stat / all
NAME        N    MEAN        ST. DEV      VARIANCE     MINIMUM      MAXIMUM
Y            63   577.76      520.25     0.27066E+06   119.00       3080.0
X2           63  0.53115E+06 0.42659E+06 0.18198E+12   58600.      0.20910E+07
X3           63   22369.      15549.     0.24178E+09   3396.0       71046.
X4           63   15166.      13702.     0.18774E+09   1351.0       57021.
X5           63   35619.      29753.     0.88526E+09   5400.0      0.14120E+06
X6           63  0.16311E+07 0.15960E+07 0.25473E+13   2400.0      0.78022E+07
X7           63   6480.5      6099.5     0.37204E+08   730.00       35960.

|_ols y x4

 R-SQUARE =   0.5820     R-SQUARE ADJUSTED =   0.5752
VARIANCE OF THE ESTIMATE-SIGMA**2 =  0.11498E+06
STANDARD ERROR OF THE ESTIMATE-SIGMA =   339.09
SUM OF SQUARED ERRORS-SSE=  0.70137E+07
MEAN OF DEPENDENT VARIABLE =   577.76
LOG OF THE LIKELIHOOD FUNCTION = -455.431

                     ANALYSIS OF VARIANCE - FROM MEAN
                      SS         DF            MS                F
REGRESSION       0.97670E+07      1.       0.97670E+07          84.946
ERROR            0.70137E+07     61.       0.11498E+06         P-VALUE
TOTAL            0.16781E+08     62.       0.27066E+06           0.000

                     ANALYSIS OF VARIANCE - FROM ZERO
                      SS         DF            MS                F
REGRESSION       0.30797E+08      2.       0.15398E+08         133.924
ERROR            0.70137E+07     61.       0.11498E+06         P-VALUE
TOTAL            0.37811E+08     63.       0.60017E+06           0.000


VARIABLE   ESTIMATED  STANDARD   T-RATIO        PARTIAL STANDARDIZED ELASTICITY
  NAME    COEFFICIENT   ERROR      61 DF   P-VALUE CORR. COEFFICIENT  AT MEANS
X4        0.28967E-01 0.3143E-02   9.217     0.000 0.763     0.7629     0.7604
CONSTANT   138.45      64.01       2.163     0.034 0.267     0.0000     0.2396

|_test x4 = 0.03
TEST VALUE = -0.10327E-02 STD. ERROR OF TEST VALUE  0.31430E-02
T STATISTIC = -0.32856015     WITH   61 D.F.    P-VALUE= 0.74361
F STATISTIC =  0.10795177     WITH    1 AND   61 D.F.  P-VALUE= 0.74361
WALD CHI-SQUARE STATISTIC =  0.10795177     WITH    1 D.F.  P-VALUE= 0.74249
UPPER BOUND ON P-VALUE BY CHEBYCHEV INEQUALITY = 1.00000
|_stop
```

Problem 5:

```
set nowide
sample 1 25
read(are106/fish8.txt) QF QN QP PB PF YD

STAT / ALL

OLS QF PF / RESID = EHAT PREDICT = QFHAT

STOP
```

Suppose you are estimating the supply of fish where QF is the supply for fish and PF is the price of fish.

(a) State the PRF implied by the SHAZAM command file above.

$\boldsymbol{QF_t = \beta_1 + \beta_2 PF_t + e_t}$.

(b) What sign do you expect, a priori, on β_2? Explain

Positive. We expect that as the price of the commodity increases, then quantity supplied will increase as well.

(c) Construct a (i) 90% confidence interval and (ii) 95% confidence interval for β_2. Use this confidence interval to test the hypothesis that there is no relationship between *QF* and *PF*, against the alternative that there is a relationship.

<u>In general, recall that the Confidence Interval is given by</u>:

$$\boxed{P[b_2 - (t_C)(se(b_2)) \le \beta_2 \le b_2 + (t_C)(se(b_2))] = 1 - \alpha}$$

The data needed for this are:

(i)	$b_2 = 0.034$	**(from SHAZAM output)**
(ii)	$se(b2) = 0.0044$	**(from SHAZAM output)**
(iii)	$\alpha = 10\%$ and 5%	**(set by the researcher)**
(iv)	**Critical t**	**(found in t-table once "α" is determined)**

$$t_{df}^{\alpha/2}: \quad t_{23}^{0.10/2} = 1.714; \quad t_{23}^{0.05/2} = 2.069$$

<u>90%</u>: The bounds are (0.02642; 0.04165)

H_0: $\beta_2 = 0$ vs. H_A: $\beta_2 \neq 0$. $\beta_2 = 0$ does not fall within the 90% confidence interval, so we can reject the hypothesis that *x* has no effect on *y*.

<u>95%</u>: The bounds are (0.02484; 0.04322)

H_0: $\beta_2 = 0$ vs. H_A: $\beta_2 \neq 0$. $\beta_2 = 0$ does not fall within the 95% confidence interval, so we can reject the hypothesis that *x* has no effect on *y*.

(d) Use the t-test the hypothesis that there is no relationship between *QF* and *PF*. Test at the (i) 10 % level and (ii) 5 % level.

Use the same null hypotheses tested in part (c). The following values are required:

<u>α = 0.10</u>:

$$t_{25-2}^{0.10/2} = 1.714$$

$$\hat{t} = 7.660$$

Since $\hat{t} > t_{23}^{0.05}$ we reject H_0: $\beta_2 = 0$

<u>α = 0.05</u>:

$$t_{25-2}^{0.05/2} = 2.069$$

$$\hat{t} = 7.660$$

Since $\hat{t} > t_{23}^{0.025}$ we reject H_0: $\beta_2 = 0$

(e) Use the p-statistic the hypothesis that there is no relationship between *QF* and *PF*. Clearly explain what the p-statistic values mean using the numeric result of this regression.

Since this is a two-tailed test, we need to compare. (p/2) vs (α/2), but we can get rid of the 2, thus we compare p vs *α*. The 10 % and 5% tests are given below.

$p = 0.001 < \alpha = 0.10$. $\boxed{\text{Reject } H_0: \beta_2 = 0}$.

$p = 0.001 < \alpha = 0.05$. $\boxed{\text{Reject } H_0: \beta_2 = 0}$.

p is the exact probability of making a Type I error; α is the researchers willingness to accept a Type I error. Since p < α, the exact probability is less than the willingness to accept, we reject H_0.

(f) Use this confidence interval to test the hypothesis that β_2 is 0.04. Test at the 5% level.

For the 5% level the bounds are (0.02484; 0.04165)

$H_0: \beta_2 = 0.04$ vs. $H_A: \beta_2 \neq 0.04$.

$\beta_2 = 0.04$ does fall within the 95% confidence interval, so we cannot reject the null hypothesis.

(g) Use the t-test the hypothesis to test the hypothesis that β_2 is 0.04. Test at the 5% level.

$$t_{25-2}^{0.05/2} = 2.069$$

$$\hat{t} = -1.343$$

Since $\hat{t} < t_{23}^{0.025}$ (in absolute value terms) we cannot reject $H_0: \beta_2 = 0$

(h) Use the p-test the hypothesis to test the hypothesis that β_2 is 0.04. Test at the 5% level.

Since this is a two-tailed test, we need to compare. (p/2) vs (α/2), but we can get rid of the 2, thus we compare p vs *α*. The 10 % and 5% tests are given below.

$p = 0.19234 > \alpha = 0.05$. $\boxed{\text{Cannot Reject } H_0: \beta_2 = 0.04}$.

(i) What are the default hypothesis tests for β_1 and β_2 when reading the SHAZAM results (that is, the t-statistic results)?

On SHAZAM, and probably all computer packages, the default null hypothesis for a given parameter is:

$H_0: \beta_K = 0$ vs. $H_A: \beta_2 \neq 0$

(j) (i) Clearly define the following terms and (ii) use the regression results to state the estimated values from the regression equation above:

(i) $\sum (y_t - \bar{y})^2$

(ii) $\sum (\hat{y}_t - \bar{y})^2$

(iii) $\sum (y_t - \hat{y})^2$

ANALYSIS OF VARIANCE - FROM MEAN

	SS	DF	MS	F
REGRESSION	4.9156	1.	4.9156	58.679
ERROR	1.9268	23.	0.83772E-01	P-VALUE
TOTAL	6.8424	24.	0.28510	0.000

(i) $\sum (y_t - \bar{y})^2 = 6.8424$ **(SST; Total)**

(ii) $\sum (\hat{y}_t - \bar{y})^2 = 4.9516$ **(SSR; Explained/Regression)**

(iii) $\sum (y_t - \hat{y})^2 = 1.926$ **(SSE; Error/Unexplained)**

(j) Use the information given the ANOVA table to verify that $r^2 = 0.7184$. What aspect of the estimated model is the r^2 characterizing?

r^2 = SSR/SST = 4.916/6.8424 = 0.7184. (See notes and text for r^2 discussion.)

(k) Explain the difference in the plots *QF* vs *PF* and *QFHAT* vs *PF*.

***QF* vs *PF*—this is simply plotting the actual sample data.**

***QFHAT* vs *PF*—this is plotting the predicted *QF*s vs *PF*. This is a straight line, although it may not seem like it on the crude SHAZAM plot.**

SHAZAM OUTPUT

```
|_sample 1 25
|_read(are106/fish8.txt) QF QN QP PB PF YD
UNIT 88 IS NOW ASSIGNED TO: are106/fish8.txt
   6 VARIABLES AND       25 OBSERVATIONS STARTING AT OBS       1
```

```
|_STAT / ALL
```

NAME	N	MEAN	ST. DEV	VARIANCE	MINIMUM	MAXIMUM
QF	25	13.448	0.53395	0.28510	12.300	14.800
QN	25	36925.	8262.5	0.68269E+08	24402.	47873.
QP	25	0.16000	0.37417	0.14000	0.0000	1.0000
PB	25	87.712	13.510	182.51	50.100	117.60
PF	25	84.768	13.298	176.84	56.000	118.00
YD	25	1934.1	327.83	0.10747E+06	1513.0	2610.0

```
|_OLS QF PF

 R-SQUARE =   0.7184     R-SQUARE ADJUSTED =   0.7062
VARIANCE OF THE ESTIMATE-SIGMA**2 =  0.83772E-01
STANDARD ERROR OF THE ESTIMATE-SIGMA =  0.28943
SUM OF SQUARED ERRORS-SSE=   1.9268
MEAN OF DEPENDENT VARIABLE =   13.448
LOG OF THE LIKELIHOOD FUNCTION = -3.43550
```

ANALYSIS OF VARIANCE - FROM MEAN

	SS	DF	MS	F
REGRESSION	4.9156	1.	4.9156	58.679
ERROR	1.9268	23.	0.83772E-01	P-VALUE
TOTAL	6.8424	24.	0.28510	0.000

VARIABLE NAME	ESTIMATED COEFFICIENT	STANDARD ERROR	T-RATIO 23 DF	P-VALUE	PARTIAL CORR.	STANDARDIZED COEFFICIENT	ELASTICITY AT MEANS
PF	0.34033E-01	0.4443E-02	7.660	0.000	0.848	0.8476	0.2145
CONSTANT	10.563	0.3810	27.72	0.000	0.985	0.0000	0.7855

```
|_STOP
```

|_PLOT QF PF

```
                 *=QF
                 M=MULTIPLE POINT
   15.000       |
   14.842       |
   14.684       |                                    *
   14.526       |
   14.368       |
   14.211       |
   14.053       |                              *
   13.895       |                        *  *  *
   13.737       |                *
   13.579       |                  *  ** M    *
   13.421       |                  *    *
   13.263       |                    **
   13.105       |                     M
   12.947       |                *
   12.789       |       *        * M
   12.632       |
   12.474       |
   12.316       |
   12.158       |           *
   12.000       |
                 ________________________________________
             40.000    60.000    80.000   100.000   120.000

                              PF
```

|_PLOT QFHAT PF

```
                 *=QFHAT
                 M=MULTIPLE POINT
   15.000       |
   14.842       |
   14.684       |
   14.526       |                                    *
   14.368       |
   14.211       |                              *
   14.053       |
   13.895       |                            **
   13.737       |                          *
   13.579       |                        M
   13.421       |                     M**
   13.263       |                   *M
   13.105       |                  M
   12.947       |                M
   12.789       |
   12.632       |           *
   12.474       |
   12.316       |       *
   12.158       |
   12.000       |
                 ________________________________________
             40.000    60.000    80.000   100.000   120.000

                              PF
```